

# Subdominant Lyapunov Exponents Predict Training Dynamics\*

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This project investigates the stability and convergence properties of stochastic gradient descent (SGD) through the lens of Lyapunov exponents. Using a controlled class of commuting quadratic Hessians, the study analytically and computationally verifies that the subdominant (negative) Lyapunov exponents govern the rate of re-attraction to the training manifold, while the dominant exponent determines the system’s edge of stability. The Benettin–Wolf QR method was implemented to estimate the Lyapunov spectrum online, demonstrating strong agreement with theoretical predictions under both fixed and adaptive step-size schedules. By tuning the learning rate to maintain the dominant exponent near zero, the results show clear empirical evidence that edge-stability control yields predictable convergence behavior. These findings provide a rigorous foundation for understanding stability in deep-learning dynamics and motivate future extensions to non-commuting Hessians, alternative optimizers, and noisy gradient updates under the Multiplicative Ergodic Theorem framework.

## I. INTRODUCTION

Recent empirical and theoretical work has shown that full-batch gradient descent on neural networks often enters an “edge-of-stability” (EOS) regime, where the maximum Hessian eigenvalue (sharpness) hovers near the classical linear stability threshold  $\frac{2}{\eta}$ , while the loss still decreases on long time scales. One of the main insights I build on is the work of Cohen [1] who examines a full-batch GD on real nets with a fixed step size  $\eta$ . As training starts, the sharpness  $S(\theta) \stackrel{\text{def}}{=} \lambda_{\max}(\nabla^2 L(\theta))$  increases along the GD trajectory until it approaches the classical stability cutoff of  $\frac{2}{\eta}$ . There is then a hovering near  $\frac{2}{\eta}$  for a long time horizon where the loss trends downward with local non monotonic oscillations in short time horizons.

Extensive followup work is then built on this concept. Arora [2] shows that for certain modified GD variants i.e. normalized GD and GD on  $\sqrt{L - L_{\min}}$ , one can prove the dynamics enter this EOS regime where sharpness again stabilizes near this threshold  $\frac{2}{\eta}$ . Later Damian [3] shows that a self-stabilization mechanism exists when GD with  $\eta$  becomes locally linearly unstable in the top Hessian eigen-direction and a cubic term that exists in the Taylor expansion of the loss pushes the curvature until there is a return to the stability boundary (i.e.  $S(\theta) \leq \frac{2}{\eta}$ ). Finally Chemnitz [4] observes which minima are possible attractors for SGD while simultaneously examining that some minima are in fact dynamically unstable and unreachable by SGD.

My work builds on these in quite a natural way. In particular, I hypothesize that, when the dominant Lyapunov exponents of the weight flow are controlled near zero - via adaptive learning-rate tuning or injected perturbations - the remaining more negative subdominant exponents will provide almost-sure re-attraction to the training manifold. It is important to observe that the

previous authors, in the global setting, kept the learning fixed while the curvature of the loss evolved which allowed for the dynamics to sharpen followed by a self-stabilization mechanism that constraints the trajectories to remaining near  $S(\theta) \approx \frac{2}{\eta}$ . My analysis, on the other hand, is local and linearized: I approximate the late-time dynamics in a fixed metastable region by a random Jacobian model with a stationary eigenvalue law. In this surrogate, the curvature distribution does not adapt to  $\eta$ , namely by solving  $\lambda_{i_*} \approx 0$  for the dominant Lyapunov exponent.

The Lyapunov exponent I study remains analogous to the EOS constraint, but I use it as a hypothesis on a frozen local model rather than an emergent property of the full non-linear training dynamics. Crucially, one should observe that I am not trying to answer the question of why training ends up at a certain place. Instead, I want to assume that I am already in a regime where the linearized dynamics around current parameters have a stationary curvature law, a dominant direction, and a step-size schedule that places that direction at the edge. This assumption allows me to look at the entire spectrum of subdominant exponents and see how fast their perturbations die out. In addition, I can explicitly find rates of re-attraction to the training manifold and properly characterize this entire spectrum of exponents.

I will, in the following sections, prove this hypothesis developed through these strong insights by characterizing the entire stability spectrum. I prove that curvature ordering on the increasing branch of  $x \mapsto \log|1 - \eta x|$  forces all curved subdominant modes to have strictly negative Lyapunov exponents, with quantitative finite-time contraction rates that yield explicit “re-attraction envelopes” back to the training manifold.

To complement this analysis, I estimate the top- $k$  Lyapunov exponents online during training using a Benettin–Wolf QR scheme, while the dominant exponent is held near zero via mild adaptive step-size control. By systematically varying learning-rate schedules and model hyperparameters, I can empirically assess how the Lyapunov

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spectrum governs noise amplification, recovery from perturbations, and ultimately the generalization behavior of SGD at matched training loss. In this way, I recast the edge-of-stability phenomenon into a concrete, mode-resolved stability principle: *edge alignment of the dominant exponent yields robust, quantifiable re-attraction rates determined by the subdominant Lyapunov spectrum.*

## II. SETUP

My goal is to quantify how the spectrum of local contraction/expansion rates under (S)GD governs small perturbation dynamics and re-attraction to a training manifold, while keeping the dominant rate near zero (edge-of-stability control). In realistic networks, curvature varies with both data and state, Jacobians do not commute, and closed-form Lyapunov exponents are unavailable; one must invoke the full random co-cycle machinery to even define the exponents rigorously. However, in this particular case, I can instead analyze a commuting quadratic class in which the following hold true:

- i) Per-sample Hessians share a common eigen-basis (all Jacobians commute).
- ii) Overparameterization appears as exact flat directions.
- iii) (S)GD error dynamics decouple coordinate-wise in that basis.

Later, I will provide a relevant remark in which I detail a way in which one could properly analyze the more general case, but for the scope of this project and a working example, I will work solely inside of this class. Within this class, I can easily do the following:

- i) I can write down the update exactly
- ii) I can define the training manifold explicitly
- iii) Compute the quantities of interest by elementary law-of-large arguments

I consider a probability space  $\mathcal{P} = (\Omega, \mathcal{F}, \mathbb{P})$ , and a measurable space of data (or mini-batch indices)  $\mathcal{S} = (\Xi, \mathcal{X})$ . A *data process* is a sequence of random variables  $(\xi_t)_{t \geq 0}$ ,  $\xi_t : \mathcal{P} \rightarrow \mathcal{S}$ , each taking values in  $\Xi$ . Throughout, I assume  $(\xi_t)_{t \geq 0}$  is *independent and identically distributed* with *common law*  $P_\xi := \mathbb{P} \circ \xi_t^{-1}$ , that is, the probability measure on  $(\Xi, \mathcal{X})$  induced by any  $\xi_t$  under  $\mathbb{P}$ .

I let  $P \in \mathbb{N}$  be the ambient parameter dimension and let  $\langle x, y \rangle \stackrel{\text{def}}{=} x^\top y$  denote the Euclidean inner product on  $\mathbb{R}^P$  with induced norm  $\|x\| = \sqrt{\langle x, x \rangle}$ . A parameter vector is denoted  $w \in \mathbb{R}^P$  and  $w^* \in \mathbb{R}^P$  is distinguished reference parameter (the "interpolating" solution for the

class I introduce). For each  $\xi \in \Xi$  I consider a per-sample quadratic loss:

$$\ell_\xi(w) = \frac{1}{2}(w - w^*)^\top H_\xi(w - w^*) \quad (1)$$

where  $H_\xi \in \mathbb{R}^{P \times P}$  is symmetric and positive semi-definite. The defining structural hypothesis of the present class is simultaneous diagonalizability of the family  $\{H_\xi : \xi \in \Xi\}$ : i.e. there exists an orthogonal matrix  $U \in \mathbb{R}^{P \times P}$  s.t.  $\forall \xi \in \Xi$ :

$$H_\xi = U \Lambda_\xi U^\top, \quad \Lambda_\xi = \text{diag}(\lambda_1(\xi), \dots, \lambda_P(\xi)), \quad \lambda_i(\xi) \geq 0 \quad (2)$$

Equivalently, all  $H_\xi$  commute pairwise and share the orthonormal eigen-basis given by the columns  $u_1, \dots, u_P \in U$ . This hypothesis captures a tractable "commuting curvature" regime in which the stochastic gradient dynamics decouple coordinate-wise in the spectral basis.

Overparameterization is encoded by the flat index set:

$$\mathcal{Z} \stackrel{\text{def}}{=} \left\{ i \in \{1, \dots, P\} : \lambda_i(\xi) = 0 \text{ for } P_\xi - \text{almost every } \xi \right\} \quad (3)$$

Along these directions the loss is flat for every sample. The corresponding training manifold is the affine subspace:

$$\begin{aligned} \mathcal{M} &\stackrel{\text{def}}{=} \left\{ w \in \mathbb{R}^P : \prod_{\mathcal{Z}^c} U^\top(w - w^*) = 0 \right\} \\ &= w^* + \text{span}\{u_i : i \in \mathcal{Z}\} \end{aligned} \quad (4)$$

where  $\prod_{\mathcal{Z}^c}$  denotes the coordinate projection that retains the  $\mathcal{Z}^c$  components and zeros the  $\mathcal{Z}$  components. For affine subspaces, the tangent space  $T_{w^*}$  is the same at every point and equals the underlying subspace:

$$T_{w^*} \mathcal{M} = \text{span}\{u_i : i \in \mathcal{Z}\} \quad (5)$$

i.e. the vectors  $u_i$  with  $i \in \mathcal{Z}$  are exactly the "flat" directions – moving a little in any of those directions keeps you on  $\mathcal{M}$ . Because I defined  $\mathcal{M}$  via linear constraints, I can let  $A \stackrel{\text{def}}{=} \prod_{\mathcal{Z}^c} U^\top \implies \mathcal{M} = \{w : A(w - w^*) = 0\}$ . For a linear constraints, the tangent space at  $w^*$  is the kernel of  $A$ :  $T_{w^*} \mathcal{M} = \ker A = \text{span}\{u_i : i \in \mathcal{Z}\}$  therefore I can conclude that dually, the orthogonal complement at  $w^*$  is  $(T_{w^*} \mathcal{M})^\top = \text{range}(A^\top) = \text{span}\{u_i : i \in \mathcal{Z}^c\}$ .

Given a step-size sequence  $(\eta_t)_{t \geq 0} \subset (0, \infty)$ , the stochastic gradient descent (SGD) update at time  $t$  is the mapping:

$$\Phi_t : \mathbb{R}^P \rightarrow \mathbb{R}^P, \quad \Phi_t(w) = w - \eta_t \nabla \ell_{\xi_t}(w) \quad (6)$$

Since  $\nabla \ell_\xi(w) = H_\xi(w - w^*)$ , it is natural to study the error iterate  $e_t \stackrel{\text{def}}{=} w_t - w^*$ , which evolves according to the linear relation:

$$e_{t+1} = (I - \eta_t H_{\xi_t}) e_t \quad (7)$$

I can then write a one-step linear factor which is the linear map that carries a small parameter perturbation forward by one iterate:

$$J_t(w) \stackrel{\text{def}}{=} D_w \Phi_t(w) = I - \eta_t \nabla^2 \ell_{\xi_t}(w) \quad (8)$$

For a small perturbation  $\delta w_t$  around  $w_t$ , the next perturbation satisfies  $\delta w_{t+1} = J_t(w_t) \delta w_t + o(\|\delta w_t\|)$ . However, in our particular quadratic class defined in (2), I have that  $\ell_{\xi}(w) = \frac{1}{2}(w - w^*)^\top H_{\xi}(w - w^*)$  so  $\nabla^2 \ell_{\xi}(w) = H_{\xi}$  is constant; which means I have:

$$J_t = I - \eta_t H_{\xi_t} \quad (9)$$

is exact i.e. (no  $o(\cdot)$  term). With the shared eigen-basis  $U$ , this diagonalizes as:

$$J_t = U \text{diag}(m_{t,1}, \dots, m_{t,P}) U^\top, \quad m_{t,i} = 1 - \eta_t \lambda_i(\xi_t) \quad (10)$$

this linear factor is the driving force behind why this works. Geometrically,  $J_t$  is the linearization of  $\Phi_t$  which describes the dynamics of moving along the parameter space. This means it is the linear map that best explains how nearby points move relative to you. In flat directions (tangent to  $\mathcal{M}$ ),  $J_t$  provides neutral drift. In curved directions (normal to  $\mathcal{M}$ ),  $J_t$  damps or amplifies depending on curvature and step size. The product of these linearizations over time is the mechanism by which local geometry aggregates into global stability/instability measurements. Thus, operationally this linear factor encodes stability per direction. In eigen-direction  $u_i$ , the scalar multiplier is  $m_{t,i}$ . If  $|m_{t,i}| < 1$  on average, perturbations shrink in that direction (a negative Lyapunov exponent), if  $|m_{t,i}| > 1$ , they grow (positive Lyapunov exponent).

Introducing spectral coordinates  $z_t \stackrel{\text{def}}{=} U^\top e_t \in \mathbb{R}^P$ , the dynamics decouple into the coordinate-wise scalar recursions:

$$z_{t+1,i} = m_{t,i} z_{t,i}, \quad (i = 1, \dots, P; t = 0, 1, 2, \dots) \quad (11)$$

Hence, for each  $i$ , one has the explicit product representation:

$$z_{t,i} = \left( \prod_{s=0}^{t-1} m_{s,i} \right) z_{0,i} \quad (12)$$

Two orthogonal projectors will be used repeatedly. The projector onto the manifold tangent and the projector onto the normal complement respectively are:

$$P_{\parallel} \stackrel{\text{def}}{=} U \Pi_{\mathcal{Z}} U^\top \quad (13)$$

$$P_{\perp} \stackrel{\text{def}}{=} U \Pi_{\mathcal{Z}^c} U^\top \quad (14)$$

The orthogonal distance from a point  $w$  to the manifold  $\mathcal{M}$  is then:

$$\begin{aligned} \text{dist}_{\perp}(w, \mathcal{M}) &\stackrel{\text{def}}{=} \|P_{\perp}(w - w^*)\| \\ &= \|\Pi_{\mathcal{Z}^c} U^\top (w - w^*)\| \\ &= \|z_{\mathcal{Z}^c}\| \end{aligned} \quad (15)$$

Finally, I want to introduce some mild regularity that will be used throughout the following arguments. The step sizes are assumed uniformly bounded and positive, i.e., there exist constants  $0 < \underline{\eta} < \bar{\eta} < \infty$  with  $\underline{\eta} \leq \eta_t \leq \bar{\eta}$  for all  $t$ . I assume basic integrability of the curvature, for instance  $\mathbb{E} \|H_{\xi}\|_{\text{op}} < \infty$ . Whenever logarithmic time-averages are formed, I additionally require the standard “stability window” ensuring finiteness of  $\log |m_{t,i}|$ ; a sufficient condition is  $\eta_t \lambda_i(\eta_t) < 2$  almost surely for all  $i$  and  $t$ , which guarantees  $|m_{t,i}| < 1$  in the curved directions and  $\log^+ |m_{t,i}| \in L^1(\mathbb{P})$ . Under these hypotheses, the class is fully specified and all subsequent quantities are well defined without having to fully invoke the random-cocycle / multiplicative-ergodic machinery.

### III. COMPUTING THE LYAPUNOV EXPONENTS IN $H_{\xi}$

In Section II, I defined a class of per-sample quadratic losses (1) with simultaneously diagonalizable Hessians (2), an affine manifold of global minimizers  $\mathcal{M}$  (4), and the stochastic gradient map  $\Phi_t$  (6) whose one-step linear factor was (10). In this section—equipped with these tools—I compute the Lyapunov exponents exactly. This is a consequence of our class choice, which yields  $J_t$  that commute and share the fixed eigen-basis  $U$ . The dynamics decouple coordinate-wise into products of scalars  $\{m_{t,i}\}$ , enabling a direct law-of-large-numbers/ergodic argument (no cocycle machinery is needed in this setting; cf. Birkhoff’s theorem [5]).

I restate the standing assumptions:

- (i)  $(\xi_t)_{t \geq 0}$  is i.i.d. with common law  $P_{\xi}$  on  $(\Xi, \mathcal{X})$ ;
- (ii) the stepsizes  $(\eta_t)_{t \geq 0} \subset (0, \infty)$  are bounded: there exist  $0 < \underline{\eta} \leq \bar{\eta} < \infty$  such that  $\underline{\eta} \leq \eta_t \leq \bar{\eta}$  for all  $t$ ;
- (iii) for each  $i \in \{1, \dots, P\}$ , the log-integrability window holds:

$$\log^+ |1 - \eta_t \lambda_i(\xi_t)| \in L^1(\mathbb{P}) \quad \text{for all } t,$$

which is guaranteed, for example, if  $0 \leq \eta_t \lambda_i(\xi_t) \leq 2 - \varepsilon$  a.s. for some  $\varepsilon > 0$ .

**Remark 1** (Further Restriction to Subcritical Window). *Whenever logarithmic time averages are formed, I additionally require the standard “stability window” ensuring finiteness of  $\log |m_{t,i}|$ ; a sufficient condition is  $0 < \eta_t \lambda_i(\xi_t) < 2$  almost surely for all  $i$  and  $t$ , which guarantees  $|m_{t,i}| < 1$  in the curved directions and  $\log^+ |m_{t,i}| \in L^1(\mathbb{P})$ . Under these hypotheses, the class is fully specified and all subsequent quantities are well defined.*

*For the comparison results in Section IV I will sometimes strengthen this window and assume that, for the subset of curved modes under comparison, there exist constants  $1 < \alpha \leq \beta < 2$  such that*

$$\alpha \leq \eta_t \lambda_i(\xi_t) \leq \beta \quad \text{a.s. } \forall \text{ such } i, t.$$

On this compact subinterval of  $(1, 2)$ , the map  $x \mapsto \log|1 - \eta_t x|$  is  $C^1$  and strictly increasing, so curvature ordering passes monotonically through  $f_{\eta_t}$ .

With these assumptions, define the error  $e_t \stackrel{\text{def}}{=} w_t - w^*$  and its spectral coordinates  $z_t \stackrel{\text{def}}{=} U^\top e_t$ . For each coordinate  $i \in \{1, \dots, P\}$  the dynamics decouple as

$$z_{t+1,i} = m_{t,i} z_{t,i}, \quad z_{t,i} = \left( \prod_{s=0}^{t-1} m_{s,i} \right) z_{0,i}, \quad t \geq 1, \quad (16)$$

and the almost-sure limit

$$\lambda_i \stackrel{\text{def}}{=} \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=0}^{T-1} \log |m_{t,i}| \quad (17)$$

exists. The following equivalent characterizations hold:

- (i) **Directional growth.** For the eigenvector  $u_i$  (the  $i$ -th column of  $U$ ),

$$\lim_{T \rightarrow \infty} \frac{1}{T} \log \|J_{T-1} \cdots J_0 u_i\| = \lambda_i \quad \text{a.s.} \quad (18)$$

- (ii) **Singular-value growth.** Let  $\sigma_1(\Pi_T) \geq \dots \geq \sigma_P(\Pi_T)$  denote the singular values of  $\Pi_T \stackrel{\text{def}}{=} J_{T-1} \cdots J_0$ . Then, as a multiset,

$$\left\{ \lim_{T \rightarrow \infty} \frac{1}{T} \log \sigma_j(\Pi_T) \right\}_{j=1}^P = \{\lambda_1, \dots, \lambda_P\}. \quad (19)$$

I can say this because each  $J_t$  is symmetric and the  $J_t$  commute,  $\prod_T$  is symmetric and orthogonally diagonalizable, hence its singular values are the absolute values of its eigenvalues  $|\prod_t m_{t,i}|$ .

- (iii) **Closed forms for i.i.d. data.**

– If  $\eta_t \equiv \eta$ , then

$$\lambda_i = \mathbb{E}[\log |1 - \eta \lambda_i(\xi)|]. \quad (20)$$

– If  $\eta_t$  is bounded deterministic, then

$$\lambda_i = \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}[\log |1 - \eta_t \lambda_i(\xi)|], \quad (21)$$

and if  $\eta_t \rightarrow \bar{\eta}$  with uniform integrability (e.g. under the same stability window), then

$$\lambda_i = \mathbb{E}[\log |1 - \bar{\eta} \lambda_i(\xi)|].$$

Moreover, for every flat index  $i$  (cf. (3)) with  $\lambda_i(\xi) \equiv 0$  a.s., one has  $m_{t,i} \equiv 1$  and hence  $\lambda_i = 0$ ; these directions span  $\mathbb{T}_w^* \mathcal{M}$ .

I prove (16)–(21), starting with (16). Because  $J_t$  is diagonal in the fixed basis  $U$ , I write  $\Pi_T \stackrel{\text{def}}{=} J_{T-1} \cdots J_0$ , I have:

$$\Pi_T = U \text{diag} \left( \prod_{t=0}^{T-1} m_{t,1}, \dots, \prod_{t=0}^{T-1} m_{t,P} \right) U^\top \quad (22)$$

equivalently, in spectral coordinates  $z_t \stackrel{\text{def}}{=} U^\top (w_t - w^*)$ :

$$z_{t+1,i} = m_{t,i} z_{t,i} \implies z_{T,i} = \left( \prod_{t=0}^{T-1} m_{t,i} \right) z_{0,i} \quad (23)$$

which yields the recursion in (16). Taking logs, I get:

$$\frac{1}{T} \log \frac{|z_{T,i}|}{|z_{0,i}|} = \frac{1}{T} \sum_{t=0}^{T-1} \log |m_{t,i}| \quad (24)$$

*Constant stepsize (Birkhoff).* Fix  $i$ . With  $\eta_t \equiv \eta$ , define

$$\psi_t(\omega) := \log |m_{t,i}(\omega)| = \log |1 - \eta \lambda_i(\xi_t(\omega))| = f_i(T^t \omega),$$

where  $(\Omega, \mathcal{F}, \mathbb{P}, T)$  is the left shift on  $\Xi^\mathbb{N}$  (measure-preserving and ergodic for i.i.d. coordinates) and  $f_i \in L^1(\mathbb{P})$  by (III). By Birkhoff's ergodic theorem [5, Thm. 1.14],

$$\frac{1}{T} \sum_{t=0}^{T-1} \psi_t \rightarrow \mathbb{E}[f_i] = \mathbb{E}[\log |1 - \eta \lambda_i(\xi)|] \quad \text{a.s.} \quad (25)$$

I then of course take the limit of (24) using  $U^\top u_i = e_i$  to get:

$$\Pi_T u_i = U \text{diag} \left( \prod_t m_{t,i}, \dots, \prod_t m_{t,P} \right) e_i = \left( \prod_{t=0}^{T-1} m_{t,i} \right) u_i \quad (26)$$

which gives us:

$$\frac{1}{T} \log \|\Pi_T u_i\| = \frac{1}{T} \sum_{t=0}^{T-1} \log |m_{t,i}| \xrightarrow{\text{a.s.}} \lambda_i \quad (27)$$

this of course proves (18). Applying (25) gives us precisely the statement:

$$\lambda_i \stackrel{\text{def}}{=} \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=0}^{T-1} \log |m_{t,i}| \stackrel{\text{a.s.}}{=} \mathbb{E}[\log |1 - \eta \lambda_i(\xi)|] \quad (28)$$

which proves both (17) and (20).

*Bounded deterministic stepsizes (Kolmogorov SLLN).* Now let  $(\eta_t)$  be deterministic and bounded. Define

$$\psi_t \stackrel{\text{def}}{=} \log |1 - \eta_t \lambda_i(\xi_t)|.$$

Under the stability window in (III) there exists  $\varepsilon \in (0, 1)$  with  $|1 - \eta_t \lambda_i(\xi_t)| \in [\varepsilon, 1]$  a.s. for all  $t$ , hence  $|\psi_t| \leq |\log \varepsilon| =: C$  a.s. for all  $t$ . Let  $X_t \stackrel{\text{def}}{=} \psi_t - \mathbb{E}[\psi_t]$ . Then  $\{X_t\}$  are independent, mean-zero, and  $\text{Var}(X_t) \leq C^2$ . Since

$$\sum_{t=1}^{\infty} \frac{\text{Var}(X_t)}{t^2} \leq \sum_{t=1}^{\infty} \frac{C^2}{t^2} < \infty,$$

Kolmogorov's strong law for independent, non-identically distributed variables applies (see, e.g., [6, Thm. 2.3.10] or [7, Ch. 7]), giving

$$\frac{1}{T} \sum_{t=0}^{T-1} (\psi_t - \mathbb{E}[\psi_t]) \xrightarrow{\text{a.s.}} 0.$$

Hence

$$\frac{1}{T} \sum_{t=0}^{T-1} \log |m_{t,i}| = \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}[\log |1 - \eta_t \lambda_i(\xi)|] + o_{\text{a.s.}}(1).$$

Recalling the definition (17) of  $\lambda_i$  via time averages, I obtain the identity (21).

*Asymptotically constant stepsizes (dominated convergence + Toeplitz).* If  $\eta_t \rightarrow \bar{\eta}$  and the same stability window holds, define  $g_{\eta}(x) := \log |1 - \eta x|$ . Then  $|g_{\eta_t}(\lambda_i(\xi))| \leq |\log \varepsilon|$  and  $g_{\eta_t}(\lambda_i(\xi)) \rightarrow g_{\bar{\eta}}(\lambda_i(\xi))$  pointwise, so by dominated convergence,

$$\mathbb{E}[g_{\eta_t}(\lambda_i(\xi))] \rightarrow \mathbb{E}[g_{\bar{\eta}}(\lambda_i(\xi))] = \mathbb{E}[\log |1 - \bar{\eta} \lambda_i(\xi)|].$$

By Toeplitz's (Cesàro) lemma (see, e.g., [8, § XIII.6]), from  $a_t \rightarrow a$  I get  $\frac{1}{T} \sum_{t=0}^{T-1} a_t \rightarrow a$ ; applying this to  $a_t := \mathbb{E}[g_{\eta_t}(\lambda_i(\xi))]$  and combining with (21) yields the stated limit with  $\bar{\eta}$ .

$$\lambda_i = \begin{cases} \mathbb{E}[\log |1 - \eta \lambda_i(\xi)|], & \text{if } \eta_t \equiv \eta, \\ \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}[\log |1 - \eta_t \lambda_i(\xi)|], & \text{if } (\eta_t) \text{ is bounded deterministic} \end{cases} \quad (29)$$

Now, I can apply the work of [1] to use edge control to imply negative subdominant exponents. I start by fixing an index  $i_* \notin \mathcal{Z}$  s.t.  $\lambda_{i_*}(\xi) \geq \lambda_j(\xi)$  a.s. for all  $j \notin \mathcal{Z}$ . For constant step-size I define:

$$\varphi_i(\eta) \stackrel{\text{def}}{=} \mathbb{E}[\log |1 - \eta \lambda_i(\xi)|] \quad (30)$$

I then let  $\eta > 0$  and state a comparison lemma on the increasing branch of  $f_{\eta}$ .

**Lemma 1** (Monotone-branch comparison). *Let  $\eta > 0$  and let  $I \subset (1/\eta, 2/\eta)$  be a compact interval. Let  $X, Y$  be random variables with values in  $I$  such that  $0 \leq X \leq Y$  almost surely and  $\mathbb{P}(X < Y) > 0$ . Then*

$$\mathbb{E}[\log |1 - \eta X|] < \mathbb{E}[\log |1 - \eta Y|]. \quad (31)$$

*Proof.* For  $x \neq 1/\eta$ ,

$$f_{\eta}(x) \log |1 - \eta x|, \quad f'_{\eta}(x) = -\frac{\eta}{1 - \eta x}.$$

On  $(1/\eta, 2/\eta)$  one has  $1 - \eta x < 0$ , hence  $f'_{\eta}(x) > 0$  and  $f_{\eta}$  is strictly increasing there. Since  $X \leq Y$  almost surely and both take values in  $I \subset (1/\eta, 2/\eta)$ , I

Moreover, I am given another important consequence: flat directions give zero exponents. I see this by taking  $i \in \mathcal{Z}$  (3) which implies  $\lambda_i(\xi) \equiv 0 \implies m_{t,i} \equiv 1 \implies \lambda_i = 0$ . These span the tangent space of  $\mathcal{M}$  in neutral directions. Thus, when I tune  $\eta$  so the top expectation  $\mathbb{E} \log |1 - \eta \lambda_{\max}(\xi)| = 0$ , I am putting the dominant exponent at 0, thus all other curved directions have strictly negative exponents under mild curvature separation, implying contraction, I clarify this rigorously in the next section. Thus, I can naturally transition to our method of control that will set this dominant exponent to zero, thus giving us edge-of-stability control and allowing us to see faster contraction dictated by the negativity of the subdominant Lyapunov exponents.

#### IV. EDGE-OF-STABILITY CONTROL AND STRICT NEGATIVITY OF SUBDOMINANTS

In the last section, I was able to explicitly compute the Lyapunov exponents. Formally, I use (3), so that  $\{u_i : i \in \mathcal{Z}\}$  spans  $T_{w^*}\mathcal{M}$ . In the commuting class (2), the Lyapunov exponents of direction  $u_i$  equals:

have  $f_{\eta}(X) \leq f_{\eta}(Y)$  almost surely, with strict inequality on the event  $\{X < Y\}$ , which has positive probability. Taking expectations yields (31).  $\square$

**Remark 2** (Feasibility of  $\varphi_{i_*}(\eta) = 0$  and Cesàro edge control). *Within our logarithmic stability window  $0 \leq \eta \lambda_{i_*}(\xi) < 2$  almost surely, I have*

$$\log |1 - \eta \lambda_{i_*}(\xi)| \leq 0 \quad (31)$$

*pointwise, with equality if and only if  $|1 - \eta \lambda_{i_*}(\xi)| = 1$ , that is,  $\eta \lambda_{i_*}(\xi) \in \{0, 2\}$  almost surely. Consequently*

$$\varphi_{i_*}(\eta) \stackrel{\text{def}}{=} \mathbb{E}[\log |1 - \eta \lambda_{i_*}(\xi)|] \leq 0, \quad (32)$$

*and achieving  $\varphi_{i_*}(\eta) = 0$  for a fixed  $\eta$  typically requires a degenerate curvature distribution (for example  $\lambda_{i_*}(\xi)$  a.s. constant with  $\eta = 2/\lambda_{i_*}$ ), mirroring the deterministic quadratic edge condition  $|1 - \eta \mu_{\max}| = 1$ .*

*In more realistic stochastic settings, one enforces edge control in Cesàro average by choosing a bounded deterministic stepsize schedule  $(\eta_t)$  satisfying*

$$\lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}[\log |1 - \eta_t \lambda_{i_*}(\xi)|] = 0. \quad (33)$$

Assume in addition that there exists  $\alpha \in (1, 2)$  such that  $\alpha \leq \eta_t \lambda_i(\xi) < 2$  a.s.  $\forall$  curved indices  $i \notin Z$  and all  $t$ .

Then for each  $t$  the random variables  $\lambda_j(\xi)$  and  $\lambda_{i_*}(\xi)$  take values in some interval  $I_t \subset (1/\eta_t, 2/\eta_t)$ , so applying Lemma 1 with  $X = \lambda_j(\xi)$  and  $Y = \lambda_{i_*}(\xi)$  gives

$$\mathbb{E}[\log |1 - \eta_t \lambda_j(\xi)|] < \mathbb{E}[\log |1 - \eta_t \lambda_{i_*}(\xi)|]$$

whenever  $\mathbb{P}\{\lambda_j(\xi) < \lambda_{i_*}(\xi)\} > 0$ . Summing over  $t$ , dividing by  $T$ , and passing to the limit along (33) shows that the corresponding Cesàro exponents of all such subdominant curved modes are strictly negative.

Fix  $i_* \notin Z$  such that  $\lambda_{i_*}(\xi) \geq \lambda_j(\xi)$  almost surely for all  $j \notin Z$  (choose any maximizer in case of ties). Fix  $i_* \notin Z$  such that  $\lambda_{i_*}(\xi) \geq \lambda_j(\xi)$  almost surely for all  $j \notin Z$  (choose any maximizer in case of ties). Set  $f_\eta(x) \stackrel{\text{def}}{=} \log |1 - \eta x|$ . For  $x \neq 1/\eta$  one has

$$f'_\eta(x) = -\frac{\eta}{1 - \eta x},$$

so  $f_\eta$  is strictly decreasing on  $[0, 1/\eta)$  and strictly increasing on  $(1/\eta, 2/\eta]$ .

**Remark 3.** Before I begin our next section, I form a remark on an extremely important assumption. From Section II/III I have the exact formula  $\lambda_{i_*} = \mathbb{E}[\log |1 - \eta \lambda_{i_*}(\xi)|]$  in the constant-stepsize case (cf. (20)), and its Cesàro analogue for bounded deterministic schedules (cf. (21)). Under the uniform stability window  $0 \leq \eta \lambda_{i_*}(\xi) < 2$  a.s., the integrand satisfies  $\log |1 - \eta \lambda_{i_*}(\xi)| \leq 0$ , hence  $\lambda_{i_*} \leq 0$ . Equality  $\lambda_{i_*} = 0$  requires  $|1 - \eta \lambda_{i_*}(\xi)| = 1$  almost surely; since  $|1 - \eta x| = 1$  implies  $\eta x \in \{0, 2\}$ , the strict upper bound  $\eta \lambda_{i_*}(\xi) < 2$  forces the degenerate case  $\eta \lambda_{i_*}(\xi) = 0$  a.s. (If one instead allows the closed window  $0 \leq \eta \lambda_{i_*}(\xi) \leq 2$ , then the other endpoint  $\eta \lambda_{i_*}(\xi) = 2$  is also admissible, corresponding to tuning  $\eta = 2/\lambda_{i_*}$  in an essentially deterministic quadratic model.) Thus the statement “top exponent = 0” is a mathematically valid idealization of the edge, but in typical stochastic settings with nontrivial curvature variability the realized top exponent is slightly negative and fluctuates around 0 (a finite-time, random “wobble”); operationally one tunes  $\eta$  (or  $\eta_t$ ) so that  $\lambda_{i_*} \approx 0^-$  rather than exactly 0.

[Edge control  $\Rightarrow$  strict negativity of subdominants] Suppose  $\mathbb{E}[f_\eta(\lambda_{i_*}(\xi))] = 0$  and there exists  $\alpha \in (1, 2)$  such that

$$\alpha \leq \eta \lambda_i(\xi) < 2 \quad \text{a.s. } \forall i \notin Z.$$

Then for every  $j \notin Z$  with  $\mathbb{P}\{\lambda_j(\xi) < \lambda_{i_*}(\xi)\} > 0$ ,

$$\lambda_j = \mathbb{E}[f_\eta(\lambda_j(\xi))] < 0. \quad (34)$$

*Proof.* By the branch assumption,  $\lambda_j(\xi)$  and  $\lambda_{i_*}(\xi)$  take values in a compact interval  $I \subset (1/\eta, 2/\eta)$  on which  $f_\eta$  is strictly increasing. Hence

$$f_\eta(\lambda_j(\xi)) \leq f_\eta(\lambda_{i_*}(\xi)) \quad \text{a.s.,}$$

with strict inequality on the event  $\{\lambda_j(\xi) < \lambda_{i_*}(\xi)\}$ , which has positive probability. Taking expectations gives

$$\mathbb{E}[f_\eta(\lambda_j(\xi))] < \mathbb{E}[f_\eta(\lambda_{i_*}(\xi))] = 0.$$

But by definition  $\lambda_j = \mathbb{E}[f_\eta(\lambda_j(\xi))]$ , and I already know  $f_\eta(x) \leq 0$  on the stability window  $0 \leq \eta x < 2$ , so this strict inequality indeed means  $\lambda_j < 0$ .  $\square$

[Quantitative negative gap under separation away from  $1/\eta$ ] Assume the hypotheses of the previous paragraph and, in addition, that there exists  $1/\eta < a < b < 2/\eta$  and  $\delta > 0$  such that

$$\lambda_j(\xi), \lambda_{i_*}(\xi) \in [a, b] \quad \text{almost surely,} \quad (35)$$

and

$$\mathbb{P}\{\lambda_{i_*}(\xi) - \lambda_j(\xi) \geq \delta\} > 0. \quad (36)$$

Define

$$c \stackrel{\text{def}}{=} \inf_{x \in [a, b]} \frac{\eta}{|1 - \eta x|} > 0. \quad (37)$$

Then

$$\mathbb{E}[f_\eta(\lambda_j(\xi))] \leq -c \delta \mathbb{P}\{\lambda_{i_*}(\xi) - \lambda_j(\xi) \geq \delta\} < 0. \quad (38)$$

*Proof.* On  $[a, b] \subset (1/\eta, 2/\eta)$ , the function  $f_\eta$  is  $C^1$  and strictly increasing, with

$$|f'_\eta(x)| = \frac{\eta}{|1 - \eta x|} \geq c \quad \text{for all } x \in [a, b].$$

By the mean-value theorem, for each  $\xi$  there exists  $\theta(\xi) \in [a, b]$  between  $\lambda_j(\xi)$  and  $\lambda_{i_*}(\xi)$  such that

$$f_\eta(\lambda_j(\xi)) - f_\eta(\lambda_{i_*}(\xi)) = f'_\eta(\theta(\xi))(\lambda_j(\xi) - \lambda_{i_*}(\xi)). \quad (39)$$

On the event  $\{\lambda_{i_*}(\xi) - \lambda_j(\xi) \geq \delta\}$  I have  $\lambda_j(\xi) - \lambda_{i_*}(\xi) \leq -\delta$ , while  $f'_\eta(\theta(\xi)) \geq c > 0$  on  $[a, b]$ , so

$$f_\eta(\lambda_j(\xi)) - f_\eta(\lambda_{i_*}(\xi)) \leq -c \delta \quad \text{on } \{\lambda_{i_*} - \lambda_j \geq \delta\}.$$

Taking expectations, using  $\mathbb{E}[f_\eta(\lambda_{i_*}(\xi))] = 0$  and dropping the complement of this event, yields

$$\mathbb{E}[f_\eta(\lambda_j(\xi))] \leq -c \delta \mathbb{P}\{\lambda_{i_*}(\xi) - \lambda_j(\xi) \geq \delta\}, \quad (40)$$

which is strictly negative under the stated separation.  $\square$

This bound tells us a few things. Flat directions  $i \in Z$  satisfy  $\lambda_i = 0$  (neutral tangent dynamics). Therefore, under edge control of the dominant curved mode, all normal directions  $j \in Z^c$  are strictly contracting ( $\lambda_j < 0$ ). Thus, the slowest re-attraction rate to  $Z$  is  $\max_{j \in Z^c} \lambda_j$ . When the curvatures of the dominant and subdominant modes lie in a compact interval  $[a, b] \subset (1/\eta, 2/\eta)$ , the mean-value inequality above shows that the curvature gap  $\mathbb{E}[\lambda_{i_*}(\xi) - \lambda_j(\xi)]$  controls the magnitude of the negative Lyapunov exponent  $\lambda_j$  via a uniform slope constant  $c > 0$  on that interval.

## V. RE-ATTRACTION TO THE TRAINING MANIFOLD, WITH EXPLICIT RATE

With these tools equipped, I can now prove re-attraction to the training manifold, and provide an explicit rate for this re-attraction. This will tell us how fast, theoretically, I should converge with the subdominant exponents to the training manifold while holding the dominant exponent at  $\approx 0$ . I start by defining the normal projection and the normal distance to the manifold:

$$P_N \stackrel{\text{def}}{=} \sum_{j \in \mathcal{Z}^c} u_j u_j^\top, \quad d_t \stackrel{\text{def}}{=} \|P_N e_t\| = \left( \sum_{j \in \mathcal{Z}^c} z_{t,j}^2 \right)^{1/2} \quad (41)$$

I also have  $z_{T,j} = \left( \prod_{t=0}^{T-1} m_{t,j} \right) z_{0,j}$  from (16). I then let  $M_T \stackrel{\text{def}}{=} \max_{j \in \mathcal{Z}^c} \left| \prod_{t=0}^{T-1} m_{t,j} \right|$ . Then,  $\forall j$ , I have  $\left| \prod_{t=0}^{T-1} m_{t,j} \right| \leq M_T$ , which gives us:

$$d_T^2 = \sum_{j \in \mathcal{Z}^c} \left( \prod_{t=0}^{T-1} m_{t,j} \right)^2 z_{0,j}^2 \leq M_T^2 \left( \sum_{j \in \mathcal{Z}^c} z_{0,j}^2 \right) \quad (42)$$

so  $d_T \leq M_T \left( \sum_{j \in \mathcal{Z}^c} z_{0,j}^2 \right)^{1/2}$ . Taking logarithms, dividing by  $T$ , and passing to the limsup gives us:

$$\begin{aligned} \limsup_{T \rightarrow \infty} \frac{1}{T} \log d_T &\leq \limsup_{T \rightarrow \infty} \frac{1}{T} \log M_T \\ &\quad + \lim_{T \rightarrow \infty} \frac{1}{2T} \log \left( \sum_{j \in \mathcal{Z}^c} z_{0,j}^2 \right) \\ &= \limsup_{T \rightarrow \infty} \frac{1}{T} \log M_T \end{aligned} \quad (43)$$

because the second term is constant in  $T$ . By definition of  $M_T$ :

$$\frac{1}{T} \log M_T = \max_{j \in \mathcal{Z}^c} \lambda_j =: \lambda_*, \quad \text{a.s.} \quad (44)$$

combining the inequalities gives us:

$$\limsup_{T \rightarrow \infty} \frac{1}{T} \log d_T \leq \lambda_* \quad (45)$$

This gives us an almost-sure upper rate for any initial error  $e_0$ . I can then find a lower rate as well. Assume there exists an index  $j_* \in \mathcal{Z}^c$  such that  $\lambda_{j_*} = \lambda_* \stackrel{\text{def}}{=} \max_{k \in \mathcal{Z}^c} \lambda_k$  and  $z_{0,j_*} \neq 0$ . Let  $\Omega_*$  be the almost-sure event on which the coordinate-wise log-products converge to their Lyapunov exponents, i.e.

$$\frac{1}{T} \sum_{t=0}^{T-1} \log |m_{t,j}| \rightarrow \lambda_j \quad \text{for every } j \in \mathcal{Z}^c, \quad (46)$$

which holds with probability one by the results of III (cf. (20)–(21)). Fix  $\omega \in \Omega_*$ . Then, for  $j_*$ ,

$$\begin{aligned} \frac{1}{T} \log |z_{T,j_*}| &= \frac{1}{T} \sum_{t=0}^{T-1} \log |m_{t,j_*}| + \frac{1}{T} \log |z_{0,j_*}| \\ &\rightarrow \lambda_{j_*} = \lambda_* \end{aligned} \quad (47)$$

since  $\frac{1}{T} \log |z_{0,j_*}| \rightarrow 0$ . Because  $d_T = \|P_N e_T\| \geq |z_{T,j_*}|$ , I obtain

$$\liminf_{T \rightarrow \infty} \frac{1}{T} \log d_T \geq \lim_{T \rightarrow \infty} \frac{1}{T} \log |z_{T,j_*}| = \lambda_* \quad \text{on } \Omega_*. \quad (48)$$

Since I already established  $\limsup_{T \rightarrow \infty} \frac{1}{T} \log d_T \leq \lambda_*$  almost surely I can combine the liminf and limsup bounds which gives us

$$\lim_{T \rightarrow \infty} \frac{1}{T} \log d_T = \lambda_* \quad \text{almost surely,} \quad (49)$$

this is our lower rate. With both these rates equipped, I can write a complete finite time envelope. I have the upper envelope:

$$d_T \leq M_T \left( \sum_{j \in \mathcal{Z}^c} z_{0,j}^2 \right)^{1/2} \leq M_T \sum_{j \in \mathcal{Z}^c} |z_{0,j}|. \quad (50)$$

For the lower envelope, I observe that

$$\begin{aligned} d_T &= \left( \sum_{j \in \mathcal{Z}^c} \left| \prod_{t=0}^{T-1} m_{t,j} \right|^2 z_{0,j}^2 \right)^{1/2} \\ &\geq \max_{j \in \mathcal{Z}^c} \left| \prod_{t=0}^{T-1} m_{t,j} \right| |z_{0,j}|. \end{aligned} \quad (51)$$

Thus, I have:

$$\max_{j \in \mathcal{Z}^c} \left| \prod_{t=0}^{T-1} m_{t,j} \right| |z_{0,j}| \leq d_T \leq \left( \sum_{j \in \mathcal{Z}^c} |z_{0,j}| \right) \max_{j \in \mathcal{Z}^c} \left| \prod_{t=0}^{T-1} m_{t,j} \right| \quad (52)$$

I then can define:

$$S_T(j) \stackrel{\text{def}}{=} \frac{1}{T} \sum_{t=0}^{T-1} \log |m_{t,j}|, \quad \lambda_j \stackrel{\text{def}}{=} \lim_{T \rightarrow \infty} S_T(j), \quad (53)$$

whose a.s. convergence follows again from III (cf. (20)–(21)). Let  $\lambda_* \stackrel{\text{def}}{=} \max_{j \in \mathcal{Z}^c} \lambda_j < 0$  from IV. Since  $\max_j S_T(j) \rightarrow \lambda_*$  a.s., I have

$$\lim_{T \rightarrow \infty} \frac{1}{T} \log M_T = \max_{j \in \mathcal{Z}^c} \lambda_j =: \lambda_* \quad (54)$$

*Upper asymptotic bound.* Fix  $\varepsilon > 0$ . By (54) there exists an a.s. finite  $T_0^{(+)}(\varepsilon)$  such that for all  $T \geq T_0^{(+)}(\varepsilon)$ ,

$$M_T \leq \exp((\lambda_* + \varepsilon)T) \quad \text{a.s.}$$

Inserting into (50) yields

$$\begin{aligned} d_T &\leq \left( \sum_{j \in \mathcal{Z}^c} |z_{0,j}| \right) \exp((\lambda_* + \varepsilon)T) = C_+(\varepsilon, e_0) e^{(\lambda_* + \varepsilon)T} \\ &\quad \text{for all } T \geq T_0^{(+)}(\varepsilon) \text{ a.s.} \end{aligned} \quad (55)$$

*Lower asymptotic bound.* If there exists  $j_* \in \mathcal{Z}^c$  with  $\lambda_{j_*} = \lambda_*$  and  $z_{0,j_*} \neq 0$ , then a.s. there is  $T_0^{(-)}(\varepsilon, j_*)$  such that

$$\left| \prod_{t=0}^{T-1} m_{t,j_*} \right| \geq \exp((\lambda_* - \varepsilon)T) \quad (T \geq T_0^{(-)}(\varepsilon, j_*)).$$

By (51),

$$d_T \geq |z_{0,j_*}| e^{(\lambda_* - \varepsilon)T} = C_-(\varepsilon, e_0) e^{(\lambda_* - \varepsilon)T} \quad (56)$$

for all  $T \geq T_0^{(-)}(\varepsilon, j_*)$  a.s.,

where, e.g.,  $C_-(\varepsilon, e_0) \stackrel{\text{def}}{=} \frac{1}{2} |z_{0,j_*}|$  after increasing  $T_0^{(-)}$  if needed to absorb finite-time fluctuations.

Combining (55) and (56), I obtain: for every  $\varepsilon > 0$  there exist a.s. finite constants  $0 < C_-(\varepsilon, e_0) \leq C_+(\varepsilon, e_0) < \infty$  and  $T_0(\varepsilon, e_0)$  such that, a.s. for all  $T \geq T_0$ ,

$$C_-(\varepsilon, e_0) e^{(\lambda_* - \varepsilon)T} \leq d_T \leq C_+(\varepsilon, e_0) e^{(\lambda_* + \varepsilon)T} \quad (57)$$

*Remark (degenerate initial condition).* If  $z_{0,j} = 0$  for every  $j$  with  $\lambda_j = \lambda_*$ , then the lower rate is governed by  $\lambda_+ \stackrel{\text{def}}{=} \max\{\lambda_j : z_{0,j} \neq 0\} < \lambda_*$ ; the same argument gives  $d_T \geq \tilde{C}_-(\varepsilon, e_0) e^{(\lambda_+ - \varepsilon)T}$  a.s. for large  $T$ , while the upper envelope still involves  $\lambda_*$ .

## VI. ONLINE ESTIMATION VIA BENETTIN-WOLF QR (CONSISTENCY IN THE TOY CLASS)

I now show that the standard Benettin-Wolf (BW) QR procedure [9, 10] for online estimation of Lyapunov exponents recovers *exactly* the same exponents as the closed-form formulas obtained in our commuting-quadratic class (II-V). Recall the setting:

$$\begin{aligned} H_\xi &= U \text{diag}(\lambda_1(\xi), \dots, \lambda_P(\xi)) U^\top \\ J_t &= I - \eta_t H_{\xi_t} = U \text{diag}(m_{t,1}, \dots, m_{t,P}) U^\top, \end{aligned} \quad (58)$$

with  $m_{t,i} \stackrel{\text{def}}{=} 1 - \eta_t \lambda_i(\xi_t)$ , a fixed orthogonal  $U \in O(P)$  where:

$$O(P) = \{U \in \mathbb{R}^{P \times P} : U^\top U = U U^\top = I_P\} \quad (59)$$

and  $(\xi_t)$  i.i.d. (or stationary ergodic as in II). The directional Lyapunov exponents are the same as in (29). From here, I can formally define the classical Benettin-Wolf QR estimator. I start by fixing  $k \in \{1, \dots, P\}$ . Initialize any  $Q_0 \in O(P)$ . For  $t = 1, 2, \dots$ , define

$$Y_t \stackrel{\text{def}}{=} J_t Q_{t-1}, \quad Y_t = Q_t R_t \quad (\text{QR factorization}), \quad (60)$$

where  $Q_t \in O(P)$  and  $R_t$  is upper triangular with *positive diagonal* entries  $\{r_t(i)\}_{i=1}^P$  (this sign convention ensures

uniqueness of  $(Q_t, R_t)$ ). The BW estimates of the top  $k$  exponents at horizon  $T$  are

$$\hat{\lambda}_i(T) \stackrel{\text{def}}{=} \frac{1}{T} \sum_{t=1}^T \log r_t(i), \quad i = 1, \dots, k. \quad (61)$$

When  $k = P$  the full spectrum is estimated. From here, I can show that there is an exact consistency under the aligned initialization. I first treat the case where the QR frame is aligned with the common eigenbasis. I begin by making the following relevant assumption:

**Assumption 1** (Aligned start). *There exists a diagonal sign matrix  $S_0 = \text{diag}(\pm 1, \dots, \pm 1)$  such that  $Q_0 = U S_0$ . (Column permutations are discussed in Remark 4.)*

**Lemma 2** (One-step BW-QR under alignment). *Suppose  $Q_{t-1} = U S_{t-1}$  with  $S_{t-1} = \text{diag}(\pm 1, \dots, \pm 1)$  and set  $D_t \stackrel{\text{def}}{=} \text{diag}(m_{t,1}, \dots, m_{t,P})$ . Then*

$$Y_t = J_t Q_{t-1} = U D_t S_{t-1}, \quad (62)$$

and the (unique, positive-diagonal) QR factorization of  $Y_t$  is

$$Q_t = U S_t, \quad R_t = S_t D_t S_{t-1}, \quad (63)$$

where  $S_t \stackrel{\text{def}}{=} \text{diag}(\text{sign}(m_{t,1}), \dots, \text{sign}(m_{t,P}))$ . In particular,

$$r_t(i) = |m_{t,i}| \quad (i = 1, \dots, P). \quad (64)$$

*Proof.* Since  $U$  is orthogonal and  $D_t$  is diagonal, the columns of  $Y_t$  are orthogonal up to signs. Hence the  $Q$  factor must equal  $U$  times a diagonal sign matrix to enforce the positive-diagonal convention, and the  $R$  factor must be diagonal. Choosing  $S_t \stackrel{\text{def}}{=} \text{diag}(\text{sign}(m_{t,i}))$  yields  $R_t = S_t D_t S_{t-1}$  with positive diagonal entries  $|m_{t,i}|$ .  $\square$

**Corollary 1** (Per-step identity and limit). *Under our aligned start assumption 1, for all  $T \geq 1$  and all  $i$ ,*

$$\hat{\lambda}_i(T) = \frac{1}{T} \sum_{t=1}^T \log |m_{t,i}| \implies \hat{\lambda}_i(T) \xrightarrow[T \rightarrow \infty]{\text{a.s.}} \lambda_i, \quad (65)$$

where  $\lambda_i$  is given by (29).

*Proof.* The identity  $\log r_t(i) = \log |m_{t,i}|$  follows from Lemma 2. Almost-sure convergence of the Cesàro average is exactly the content of II for constant stepsize (Birkhoff) and bounded deterministic schedules (cf. (20)–(21)).  $\square$

I can then make the following remark about permuting the matrix and how under this permutation, the multi-set of  $\hat{\lambda} \xrightarrow[T \rightarrow \infty]{\text{a.s.}} \lambda_i$ :



**Remark 4** (Permutations). If  $Q_0 = U P_0 S_0$  with a permutation matrix  $P_0$  and signs  $S_0$ , then  $Y_t = U (P_0^\top D_t P_0) S_{t-1}$  and the same calculation shows  $Q_t = U P_0 S_t$ ,  $R_t = S_t P_0^\top D_t P_0 S_{t-1}$ ; the multi-set  $\{\hat{\lambda}_i(T)\}_{i=1}^P$  still converges a.s. to  $\{\lambda_1, \dots, \lambda_P\}$ . If one wishes to label exponents in the eigenvector order, either initialize with  $P_0 = I$  or reorder at the end.

I also can achieve consistency from arbitrary orthogonal starts. Even if  $Q_0$  is arbitrary, the commuting class allows a single data-driven alignment that preserves on-liness. This is because even if  $Q_0$  is misaligned, the commuting property guarantees that all  $J_t$  share one eigenbasis  $U$ ; therefore, after a single empirical alignment (e.g. via  $\bar{H}_M$ ), one could initialize  $Q_0 = U$  and from that point on the Benettin-Wolf QR iteration remains exactly aligned, yielding the theoretical Lyapunov exponents while keeping the algorithm fully online. I show how this is true below via the following lemma and theorem:

**Lemma 3** (One-shot alignment via an empirical mean Hessian). Let  $\bar{H}_M \stackrel{\text{def}}{=} \frac{1}{M} \sum_{t=1}^M H_{\xi_t}$ . In the commuting class,  $\bar{H}_M = U \text{diag}(\bar{\lambda}_1^{(M)}, \dots, \bar{\lambda}_P^{(M)}) U^\top$  for all  $M$ . Hence any orthonormal eigenbasis of  $\bar{H}_M$  equals  $U$  up to signed permutation. If I reset  $Q_0$  to such an eigenbasis (with positive-diagonal convention), then Lemma 2 applies from the next BW step onward.

*Proof.* Any linear combination of simultaneously diagonalizable matrices is diagonalizable by the same  $U$ . Choosing  $Q_0$  as an orthonormal eigenbasis of  $\bar{H}_M$  therefore aligns it with  $U$  up to signed permutation.  $\square$

**Theorem 1** (BW-QR consistency in the commuting-quadratic class). In the commuting-quadratic class, the Benettin-Wolf QR estimator with either

1. aligned initialization  $Q_0 = U S_0$ , or
2. arbitrary  $Q_0$  followed by the one-shot alignment in Lemma 3,

satisfies, for each  $i \in \{1, \dots, P\}$ ,

$$\hat{\lambda}_i(T) \xrightarrow[T \rightarrow \infty]{\text{a.s.}} \lambda_i, \quad (66)$$

with  $\lambda_i$  given by (29). In particular, under the edge-control conditions of Section IV the BW estimates recover the strictly negative normal exponents and hence the re-attraction rate  $-\lambda_*$  of V.

*Proof.* Case (1) is Corollary 1. For case (2), run any finite number  $M$  of steps with arbitrary  $Q_0$ , compute an orthonormal eigenbasis of  $\bar{H}_M$  to realign  $Q_0$  (Lemma 3), and continue. The finite prefix does not affect the limit of  $\frac{1}{T} \sum_{t=1}^T \log r_t(i)$ .  $\square$

Furthermore, if  $\lambda_i(\xi) \equiv 0$  a.s., then  $m_{t,i} \equiv 1$ , so  $r_t(i) = 1$  and  $\hat{\lambda}_i(T) \equiv 0$ , matching  $T_{w^*} \mathcal{M}$ . The positive-diagonal QR convention ensures  $r_t(i) = |m_{t,i}|$  even when

$m_{t,i} < 0$  (crossing  $1/\eta$ ), consistent with Lyapunov exponents defined via singular values. Re-orthonormalizing every  $p \geq 1$  steps accumulates a block-upper-triangular factor with the same diagonal products; dividing by total time leaves the limits unchanged.

In conclusion, the toy commuting class, BW-QR produces *per-step identities*  $\log r_t(i) = \log |m_{t,i}|$ , so the on-line estimates converge almost surely to the theoretical exponents (29). Under the edge-of-stability tuning of the top mode, the estimated subdominant exponents are strictly negative and coincide with the asymptotic re-attraction rate proved in V.

## VII. RESULTS

Having fully defined our commuting quadratic class and explicitly computed its Lyapunov spectrum, I now numerically verify that these theoretical exponents coincide with the values estimated via the Benettin-Wolf QR algorithm. This verification is performed under multiple deterministic and bounded step-size schedules, demonstrating that the empirical convergence rates obtained from the QR estimator match our closed-form theoretical predictions across different step-size protocols.

In addition, I investigate the edge-of-stability control regime, in which the dominant Lyapunov exponent is actively maintained near zero through learning-rate tuning. The numerical results confirm that when the leading exponent is controlled at this edge, the subdominant exponents remain strictly negative and govern exponential re-attraction to the training manifold at rates consistent with our analytical derivations.

Finally, repeating the experiments across several bounded step-size paradigms further validates the robustness of our theoretical framework: even under varied deterministic learning-rate schedules, the Benettin-Wolf estimates consistently reproduce the predicted Lyapunov exponents. These results together provide computational evidence that the theoretical expressions derived for the commuting, diagonalizable class remain accurate and predictive under practical, numerically implemented conditions, despite the non-tightness of certain assumptions (such as strict commutativity and global diagonalizability).

However, I find it relevant to repeat again that this strongly correlates with the theory due to the assumptions I have placed upon it. In fact, the SGD dynamics I consider (c.f. (7)) alone do not generalize SGD and only produce randomness multiplicatively via the sampling procedure and generation of the Hessian  $H_\xi$ . So, I take these results with the understanding that more advance tools would become necessary to generalize these results in a more nuanced way, although this does not detract from the soundness and conceptual proofs of this project.

### A. Convergence Time without Stability Control

Below I plot the time evolution of the distance to the flat manifold (cf. (15)). To monitor the confinement of the SGD trajectory, I track the normal distance

$$d_t = \|P_{\perp} e_t\|,$$

where  $e_t = w_t - w^*$  denotes the parameter error and  $P_{\perp}$  is the orthogonal projector onto the non-flat eigenspace of the Hessian. The results in Figure 1 show strong convergence of  $d_t$  to 0, which is exactly what I would expect: components along the strongly negative exponents contract very rapidly, so the fast contracting modes die out quickly and the trajectory is forced onto the flat manifold.

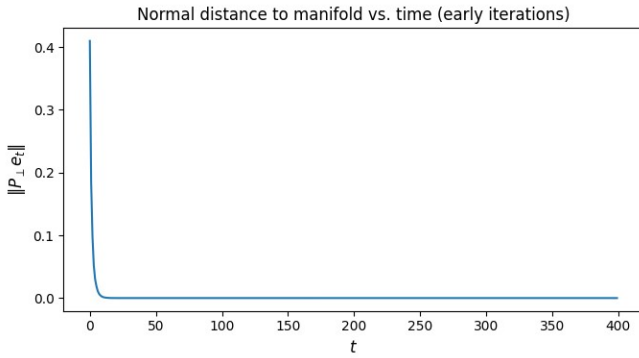


FIG. 1. Normal distance  $\|P_{\perp} e_t\|$  of the SGD parameter error  $e_t = w_t - w^*$  to the flat training manifold versus iteration  $t$ .

I see a similar phenomenon in the log-normal distance. The plateau in Figure 2 is caused by the double-precision numerical floor, where the system jitters once the fast modes have collapsed and the slowest contracting mode dominates.

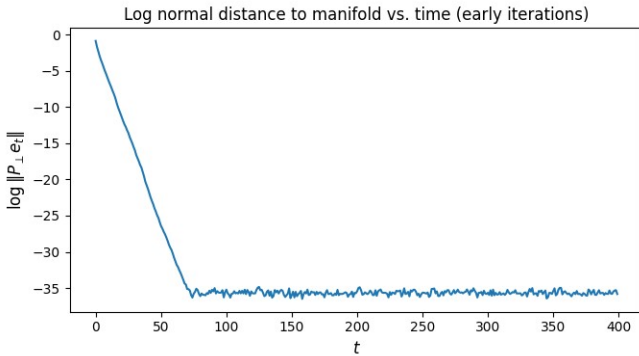


FIG. 2. Logarithm of the normal distance  $\log \|P_{\perp} e_t\|$  versus iteration  $t$ .

This behavior is also numerically verifiable. From the Lyapunov spectrum I have that the dominant normal exponent is  $\lambda_* \approx -0.39$ , and normal distances behave

asymptotically like  $d_T \approx Ce^{\lambda_* T}$  once the fastest modes have died out. Thus, when I go from  $d_0 \approx 2.1$  to the machine floor  $d_T \approx 2 \times 10^{-16}$  I obtain

$$2 \times 10^{-16} \approx 2.1e^{\lambda_* T} \implies T \approx \frac{\ln\left(\frac{2 \times 10^{-16}}{2.1}\right)}{\lambda_*} \approx \frac{-37.5}{-0.39} \approx 96$$

so I would *expect* on the order of 100 steps before convergence to the numerical floor, which matches what I observe. Finally, I report the same normal-distance diagnostic under multiple step-size schedules:

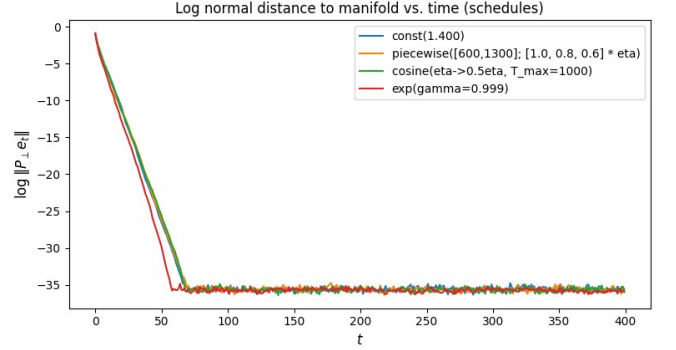


FIG. 3. Logarithm of the normal distance  $\log \|P_{\perp} e_t\|$  versus iteration  $t$  with differing step-size schedules.

### B. Theoretical Computation of Lyapunov Exponents vs Benettin–Wolf QR Estimation

Below I show an almost perfect correspondence between our Theoretical computation of Lyapunov exponents and the numerical computation of these exponents under the Benettin–Wolf QR estimation scheme:

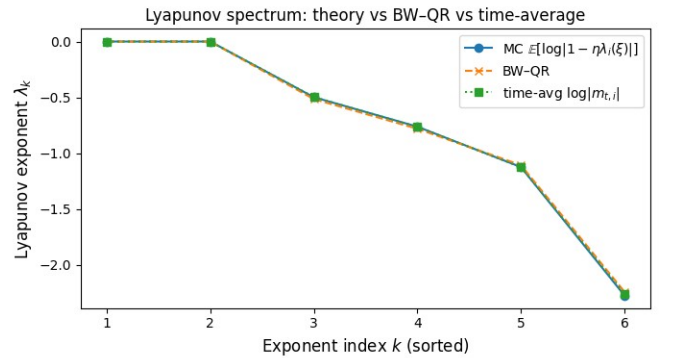


FIG. 4. Lyapunov Exponents  $\lambda_k$  for the commuting quadratic toy class, sorted so that  $\lambda_1 \geq \dots \geq \lambda_P$ . I compare Monte Carlo Evaluation of  $\mathbb{E}[\log|1 - \eta\lambda_i(\xi)|]$  (circles), Benettin–Wolf QR estimates (crosses), and time-averaged  $\log \frac{1}{T} \sum_t \log |m_{t,i}|$  (squares). The curves lie almost on top of each other; in particular, the top exponent  $\lambda_1$  and subdominant exponent  $\lambda_2$  agree across all three methods

### C. Log Normal Distance Convergence using Edge-Of-Stability Control

Given multiple  $\eta$  step-size schedules, I can show again the log-normal distance of where the return dynamics are governed by the subdominant exponents:

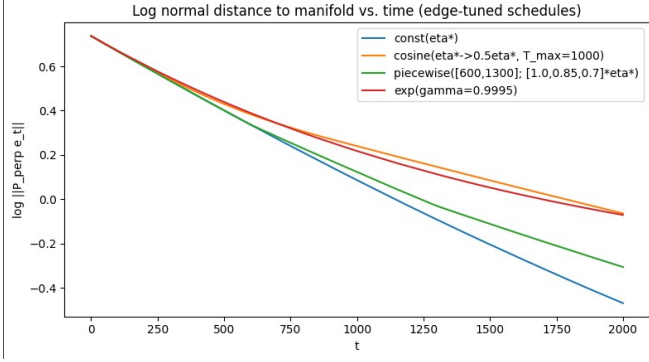


FIG. 5. Logarithm of the normal distance  $\log \|P_{\perp} e_t\|$  of the SGD parameter error  $e_t = w_t - w^*$  to the flat training manifold as a function of iteration  $t$  for several edge-tuned step-size schedules. All runs use the same commuting quadratic model and run for  $T = 2000$  iterations.

In this regime, where subdominant negative exponents govern the return dynamics to the training manifold, the system enters a regime where contraction in the normal directions is weaker than in the fully subcritical case, while tangent directions remain neutral (exponent 0). Thus, the log-normal distance  $\log \|P_{\perp} e_t\|$  decays approximately linearly with slope equal to the largest negative exponent  $\lambda_* = \max_{j \in \mathcal{Z}^c} \lambda_j < 0 \implies \log d_t \approx \lambda_* t + C$ . The results in the above graph confirm this exact prediction. The Benettin-Wolf QR estimates match the analytical expectations to numerical precision and the distance to manifold trajectories show steady exponential decay on a log scale, with rates that are accurate up to some order with the measured subdominant exponents. Under edge tuning specifically, convergence appears visually slower because the dominant contraction is intentionally neutralized, yet the relative difference in slope across schedules reveals how much more negative subdominant Lyapunov exponents yield faster re-attraction in the asymptotic sense. In conclusion, our results ultimately confirms 1) our edge control process works, 2) the rate of re-attraction is governed by the largest (least negative) remaining normal exponent  $\lambda_*$ , and 3) the measured Lyapunov spectrum provides an accurate estimate of the local stability structure of the learning dynamics.

### D. Clear Limitations and Future Work

*a. Limitations* A clear structural limitation of the present framework is the assumption that all per-sample Hessians (and hence Jacobians) commute and are simul-

taneously diagonalizable by a fixed orthogonal matrix  $U$ . In realistic networks, neither condition holds generically: the matrices  $H_{\xi}(w_t)$  depend nonlinearly on both the data and the current parameter  $w_t$ , so the family  $\{H_{\xi}(w_t)\}_{t \geq 0, \xi \in \Xi}$  typically fails to commute and does not share a common eigenbasis. Even infinitesimal perturbations of a commuting family will almost surely destroy commutativity, and non-normality or near-defective behavior can occur. As a consequence, the products  $J_{t-1} \cdots J_0$  can no longer be reduced to diagonal products of scalar multipliers  $\prod_t m_{t,i}$ ; instead, different directions are continuously mixed by the dynamics, and the growth or decay of perturbations cannot be read off from a fixed spectral basis. In particular, the clean decoupling into scalar recursions, the explicit formulas for Lyapunov exponents as time-averaged logs of  $|1 - \eta_t \lambda_i(\xi_t)|$ , and the curvature-based comparison arguments all rely crucially on this simultaneous diagonalizability and may fail dramatically once the Jacobians no longer commute. Thus explicit computation is destroyed and I require more complex machinery, which I will discuss in the future work part of this section, to utilize the same tricks to control the subdominant exponents.

A second limitation of my analysis is that all of the comparison results for Lyapunov exponents are carried out on the increasing branch of the map  $f_{\eta}(x) = \log|1 - \eta x|$ , namely  $x \in (1/\eta, 2/\eta)$ . On this branch, larger curvature yields a larger (less negative) exponent, which makes it possible to show that tuning the most curved mode to the edge ( $\lambda_{i_*} = 0$ ) forces all strictly less curved modes to have strictly negative exponents. However, this restriction discards the strongly subcritical regime  $x \in [0, 1/\eta)$ , where  $f_{\eta}$  is strictly decreasing, so that larger curvature would instead correspond to more negative exponents (stronger contraction). As a result, the current theory cannot describe mixed situations in which some spectral modes satisfy  $\eta \lambda_i(\xi) < 1$  while others satisfy  $\eta \lambda_j(\xi) > 1$ , nor can it quantify how the Lyapunov spectrum reorganizes as one moves from a safely stable regime ( $\eta \lambda \ll 1$ ) toward the edge ( $\eta \lambda \approx 2$ ). A natural improvement would be to develop a two-branch comparison theory that treats  $[0, 1/\eta)$  and  $(1/\eta, 2/\eta)$  on equal footing, allowing one to explicitly characterize how step-size choices trade off very strongly contracting directions against near-edge directions, and to design “spectral shaping” schedules that keep some modes deliberately over-damped while driving others as close to the edge as possible.

Finally, while the present results demonstrate consistent agreement between the theoretical and empirical Lyapunov spectra, further improvements can be achieved through finer parameter tuning and comparative testing. In particular, refining the learning-rate bisection to target  $\lambda_{\max}$  even closer to zero and exploring alternative step-size schedules (e.g., adaptive or cyclical variants) could yield clearer separation among subdominant modes. Incorporating controlled gradient noise would also allow direct measurement of variance amplification, validating the predicted inverse scaling with

$|\lambda_i|$ . Finally, running cross-architecture and optimizer comparisons under matched training loss would establish how these stability spectra correlate with generalization performance—completing the empirical link between edge-controlled Lyapunov dynamics and learning outcomes.

*b. Future Work* There are several promising directions in which this work could be extended. The most immediate is through the application of the *Multiplicative Ergodic Theorem* (MET), which would allow one to generalize the class of commutable, quadratic Hessians considered here to the broader and more realistic setting that stochastic gradient descent (SGD) actually produces. While such an extension would make explicit computation of the Lyapunov exponents intractable, it would enable the derivation of tight analytical bounds on their magnitudes and rates, thereby providing a rigorous framework for verifying the qualitative behavior observed in this study. This direction is also mathematically rich, as the MET requires the notion of cotypes, implying that the tangent vectors to the training manifold would naturally be represented as cotangent bundles—an abstraction that I did not yet feel sufficiently comfortable to employ in the present work.

Furthermore, extending this framework across different optimizers could reveal how convergence and stability properties vary across diverse learning regimes. It would also be valuable to incorporate additive noise into the update rule, for instance by considering an extended form

$$e_{t+1} = (I - \eta H_{\xi_t})e_t - \eta \zeta_t,$$

which captures both multiplicative and additive stochasticity. Similarly, adding *batch-sampling noise* at stages beyond the current  $\xi_t$ -sampling step would yield a more faithful representation of practical SGD while remaining theoretically consistent.

In conclusion, this project successfully achieved its goal of analytically and computationally verifying that the subdominant Lyapunov exponents govern the stability

and convergence of training dynamics in a significant way. These results motivate further theoretical generalization and experimental exploration, with the ultimate goal of developing a comprehensive theory that bridges the analytical structure of Lyapunov spectra with the practical behavior of SGD in real-world training scenarios.

## VIII. DISCLOSURE OF A.I. USAGE

I used ChatGPT 5.1 for multiple instances which I will explicitly detail here:

- 1) I used ChatGPT as a reviewer to check consistency of notation e.g. after I finished the math, I made ChatGPT go through and check to see if I had clear notational inconsistencies that didn't match what I actually meant as my notes are all over the place.
- 2) I used ChatGPT to locate certain theorems that I knew I needed after I reached a certain blockage in my proof process and it helped me locate good references which I could then pull myself and actually integrate into the proof.
- 3) I used ChatGPT in my code to do a full code audit to make sure every single line of code was exactly what I wanted to write in my paper and that I didn't miss anything. It aptly observed that I had defined my overparameterization function enforce flat Z, but never actually used it. It is unlikely without this audit I would have caught that.
- 4) I used ChatGPT to help with some of the prose I wrote. I did this by fully writing out exactly what I wanted to say, and then asking ChatGPT to clean up things like run-on sentences that would read as incoherent to the reader. I think audited the response and did a back and forth on that writing until it read exactly how I wanted to say it (without coming off as A.I. language writing which lacks inflection).

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